Morankar Kaushal Nandkumar

cs23btech11037@iith.ac.in | kaushal.n.morankar@gmail.com | +91 7498583320 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.51
Global Public School and Jr. College	Class XII (Maharashtra State Board)	2023	84.83%
New Era English School	Class X (Maharashtra State Board)	2021	97.60%

WORK EXPERIENCE

Arcesium India - *Software Engineering Intern*

- **Engineered a persistence layer for the trade consumer pipeline**, storing precomputed trade data to replace expensive database joins and serve as a dedicated data source for downstream computations; **reduced DB call latency by ~97%**
- **Redesigned high-volume trade data retrieval** to consume from the persistence table instead of reconstructing data through database joins, **reducing memory utilization by ~65%** and improving scalability of large-data processing.
- **Refactored an Arcesium-specific Java framework into a standalone executable JAR**, separating framework functionality from the core application and reducing runtime heap requirements while improving modularity and maintainability.

Inter IIT Tech Meet 14.0 - *Pathway*

- Developed an agentic workflow with dynamic tool selection, enabling the system to invoke the most relevant investigation tools based on transaction risk and contextual signals.
- Designed a research agent to continuously analyze production data, evaluate emerging detection techniques, and validate improvements through shadow-mode testing before deployment.

ACHIEVEMENTS

- Received the prestigious **KVPY Fellowship (SA)** from Govt. of India
- Secured **2nd place** in Flipr Hackathon 2025, an AI programming competition.
- Achieved a maximum rating of **1592**- **Specialist** on Codeforces.

PROJECTS

High-Throughput User-Space Market Data Feed Handler

- Architected a low-latency UDP multicast MDP3 feed handler leveraging **AF_XDP (kernel bypass)** and zero-copy eBPF redirection to achieve direct NIC-to-user-space DMA packet processing via kernel bypass.
- Built hand-rolled UAPI memory-mapped ring structures (UMEM) combined with a cache-aligned, lock-free **SPSC** ring buffer for fast thread handoff without syscalls or dynamic allocation.
- Developed SBE binary parsing and sequence gap tracking with an automated UDP simulator to validate packet loss recovery.

DSL for Graph Processing

- Designed **GraphX**, a C++ DSL for graph algorithms and operations using **C++, Flex, and Yacc**, enabling simpler syntax for declaring and managing variable scopes.
- Built core compiler components including **AST, Symbol Table, and automated build system** using **Makefile**, and implemented **semantic analysis, type/scope checking, and transpilation**.

Virtual Memory Management in gemOS

- Implemented **virtual memory management** in gemOS with VMA creation, merging, splitting, and protection.
- Implemented **lazy allocation and page-fault handling** with on-demand page allocation and page-table updates.

- Developed **Copy-on-Write (CoW) fork** using shared frames, reference counting, and CoW fault handling.

Cache Simulator

- Built a C++ cache simulator that mirrors real cache behavior and reports hits, misses, and performance stats.
- Added configurable policies: **LRU, FIFO, RANDOM** for replacement and **Write-Back, Write-Through** for writes.

Congestion-Aware Ad-Hoc Distance Vector (CA-AODV)

- Developed a custom Congestion-Aware AODV (CA-AODV) **routing** protocol in **NS-3** for Mobile Ad-hoc Networks (**MANETs**).
- Enhanced route discovery using **real-time queue depth** and **link** capacity, enabling traffic to bypass congested network paths.

RELEVANT COURSES

Computer Science: Data Structures, Algorithms, Operating Systems, Databases, Theory of Computation, Networks, Compilers, Software Engineering, Network Security, Malware Analysis, GPU Programming

AI and Mathematics: Foundations of Machine Learning, Deep Learning, NLP, Reinforcement Learning, Data Mining, Fraud Analytics, Linear Optimization, Probability and Random Variables, Linear Algebra, Statistics

POSITIONS OF RESPONSIBILITY

Tech Manager - *Office of Career Service, IITH*

- Own the development and maintenance of the institute's placement & internship portal, serving **1,500+ students**.
- Build features that streamline recruiter onboarding, student applications, and hiring workflows.

Head, Machine Learning - *Tinkerers' Lab IITH*

- Led and contributed to major projects such as Food Item detection with calorie tracking, Emotion Cause Identification etc.
- Directed a 10 member developer team, organizing hackathons and initiatives to drive campus-wide innovation.

Teaching Assistant - *CSE IIT Hyderabad*

- Served as TA for the course **Operating Systems , Data Structures**

SKILLS

Programming Languages: C/C++, Python, Java, SQL, JavaScript, Bash

Libraries: PyTorch, Numpy, Pandas, Tensorflow, Scikit-learn, Matplotlib, React, FastAPI, Langchain

Tools & Technologies: Git, Gerrit, GCP, Docker, PostgreSQL, Makefile, LaTeX, Markdown, Vim ,CUDA